

RULES AND REGULATIONS FOR THE WRITTEN EXAMS
AT THE FACULTY OF ENGINEERING AND NATURAL SCIENCES

Please read this document carefully, sign the bottom part, and include this form with your solution sheet.

- The exam duration (start-end times) will be written on your exam sheet or will be announced, you should adhere to these times.
- You cannot leave the examination room in the **first 30 minutes** of the exam.
- You are **NOT allowed** to take the exam once a student has left one of the exam rooms.
- Talking with another student during the exam for any reason (asking for an explanation, for a calculator or eraser or any other object) is **NOT allowed**.
- If you **GIVE** to or **RECEIVE** from a student any item (eraser, calculator, pen, etc.) during the exam, this will be taken as a sign of cheating.
- **TURN OFF** your CELLPHONES AND SMART DEVICES; and PLACE THEM **away from you** such that those **CANNOT BE SEEN** from outside.
- **HAVING a CELLPHONE or a SMART DEVICE** that is **VISIBLE FROM OUTSIDE** during the exam is **NOT ALLOWED**. Interacting with a cellphone in any way (EVEN IF IT IS TURNED OFF) will be taken as a sign of cheating.

In case of a sign of cheating, or if you do not obey the above rules, there will be a disciplinary action taken towards you based on the STUDENT DISCIPLINARY REGULATIONS of YÖK Items 2.5, 2.7, and 2.8.

I understand the above-mentioned rules and regulations, and I hereby accept them, and any consequences following my actions during the exam.

Student Full Name:

Student Number:

Course Code and Title:

AIN-2002 INTRODUCTION TO DATA SCIENCE

Signature:

Date:

<u>Q.1:</u>	<u>Q.2:</u>	<u>Q.3:</u>	<u>Q.4:</u>	<u>TOTAL:</u>
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Q.1) (20 pts) Please conduct necessary R code for the following expressions:

a. Create a function in R that returns the cube of a number. Use it for the value 3. **(4pts)**

b. Plot the function $f(x)=\sin(x)+\cos(x)$ for $x \in [-\pi, \pi]$. **(4pts)**

c. Simulate rolling a 6-sided die 15 times. **(4pts)**

d. Calculate the standard deviation and variance of the numbers 5, 10, 15, and 20. **(4pts)**

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MIDTERM MAKEUP EXAM

(This is a 100-minute exam)

Q.2) (20 pts) A nutritionist claims that the average calorie intake of students is 2,000 kcal/day. A random sample of 25 students shows a mean of 2,100 kcal/day with a standard deviation of 250.

Perform a one-sample t-test at 0.05 significance level.

According to the information given, please proceed with the following:

- State the null and alternative hypotheses. (5 pts)
- Conduct the t-test. (5 pts)
- Report degrees of freedom and critical t-value. (5 pts)
- Interpret the results. (5 pts)

Q.3) (20pts)

Observed	Predicted
27.5	30.400
24.3	23.800
28.2	28.300
32.6	29.800
23.8	22.700
23.8	24.000
32.9	30.599
28.8	29.600
22.7	21.500
27.7	27.099
22.7	21.500
22.7	26.400
26.2	26.200

Using the dataset temperature_predictions.csv above, which contains observed and predicted temperature values:

- Compute the Root Mean Squared Error (RMSE). (4 pts)
- Compute the coefficient of determination (R^2). (4 pts)
- Interpret both metrics. (7 pts)
- Does the model over- or under-predict at high observed temperatures? Justify your answer. (5 pts)

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Q.4) (40 pts)

Study_Hours	Exam_Score
1	8.7
1.7	9.4
2.5	11.9
3.2	14.5
3.9	12.1
4.7	19.9
5.4	24.7
6.2	36.3
6.9	36.2
7.6	29.3
8.4	43.6
9.1	43.6

Using the dataset study_vs_score.csv presented above, which shows the relationship between study hours and exam scores:

a) Identify the dependent and independent variables. (3 pts)

b) Check whether the relationship is linear or nonlinear (use a plot).? (3 pts)

c) Fit the appropriate regression model and give the equation. (10pts)

d) Interpret the slope and intercept. (10 pts)

e) Predict the exam score for a student who studied 12 hours. (4 pts)

f) Explain whether a transformation (e.g., log or square root) would improve the model. (5 pts)

g) Assess model accuracy using R^2 and RMSE. (10 pts)

Formulas

Sum of squared residuals	$SS_{\text{res}} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$
Total sum of squares	$SS_{\text{tot}} = \sum_{i=1}^n (y_i - \bar{y})^2$
Coefficient of determination (R^2)	$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$
Simple linear regression	$y = \beta_0 + \beta_1 x$
Multiple linear regression	$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p$
Exponential regression	$y = ae^{bx}$
2 nd degree Polynomial reg.	$y = \beta_0 + \beta_1 x + \beta_2 x^2$
Intercept	$\beta_0 = \bar{y} - \beta_1 \bar{x}$
Slope	$\beta_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$
Logarithmic equation	$\ln(ae^{bx}) = \ln a + bx$ $\ln(x) = y \quad \text{if and only if} \quad e^y = x$
Root Mean Square Error (RMSE)	$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$
t-statistic	$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$
Degrees of freedom	$df = n - 1$

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MIDTERM MAKEUP EXAM

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t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
Z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										